



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR
THE DEGREE OF MASTER OF SCIENCE IN ACTUARIAL
SCIENCE**

MAIN CAMPUS

MAC 817: DATA MANAGEMENT AND ANALYSIS

Date: 17th August, 2018

Time: 11.00 - 2.00pm

INSTRUCTIONS:

- Answer ANY THREE questions



Question one: 20 marks

a) Define the following terms as used in data management and analysis:

- i. Baseline data
 - ii. Meta – analysis
 - iii. Multi – stage sampling
 - iv. Quasi experimental design
 - v. Sampling error
 - vi. Systematic review
 - vii. Data reliability
 - viii. Data validity
 - ix. Data completeness
 - x. Data integrity
- (10 marks)

- b) Briefly discuss 3 major objectives of data management.
(6 marks)
- c) State 4 reasons that necessitate a data cleaning exercise.
(4 marks)

Question two: 20 marks

- a) Briefly describe a database and state 7 functions of a database management system (DBMS)
(10 marks)
- b) Define database schema describe the various levels.
(5 marks)
- c) Using a diagram illustrate a DBMS.
(5 marks)

Question three: 20 marks

- a) List five elements of a data dictionary or a define file.
(5 marks)
- b) Differentiate between two types of databases; flat files and relational databases.
(2 marks)



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b) Differentiate between two types of databases; flat files and relational databases.

(2 marks)

- c) State the purpose of the following domains and give one example of each.
- a. Special-Purpose Domains (2 marks)
 - b. Observation Classes (2 marks)
 - c. Relationships (2 marks)
 - d. Trial Design (2 marks)
- d) Explain how you would retrieve data from excel to SAS or SPSS or STATA. (5 marks)

Question four: 20 marks

- a) What kind of data would you perform the following data analysis, give an example
- i. Chi-square (2 marks)
 - ii. T-tests and (2 marks)
 - iii. Test for proportions (2 marks)
 - iv. ANOVA (2 marks)
 - v. Correlation (2 marks)
 - vi. Regression (2 marks)

Briefly describe 4 data management software giving the particular institution or industry where they are most appropriate. (8 marks)

Question five: 20 marks

- a) Define the term "Distributed database system". (2 Marks)
- b) Discuss three advantages and three disadvantages of DBMS (6 Marks)

- c) The DDBMS should appear like a centralized DBMS by providing a series of transparencies. Distinguish between the following transparencies;
- i. Replication transparency (2 Marks)
 - ii. Fragmentation transparency (2 Marks)
 - iii. Location transparency (2 Marks)
- d) Discuss the mechanisms for enforcing security in databases (6 Marks)

THE END

- c) State the purpose of the following domains and give one example of each.
- a. Special-Purpose Domains (2 marks)
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MASENO UNIVERSITY
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**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR
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MAIN CAMPUS

MAC 818: STOCHASTIC PROCESSES

Date: 16th August, 2018

Time: 11.00 - 2.00pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions



Qn.1.(a) In the discrete time case let $X_n = Z_1 + Z_2 + \dots + Z_n$ and $Z_k = \mathcal{E}_1 + \dots + \mathcal{E}_k$ where $\mathcal{E}$'s are independent and $E(\mathcal{E}_i) = 0$. Are Z 's independent? Show that X_i is a martingale.

(6 marks)

(b) Let w_t be a Brownian motion, and let a be a fixed parameter. Show that the process

$$x_i = \frac{w_i}{\sqrt{a}}$$

has the same distribution as w_i

(6 marks)

(c) Let $(S_n: n \geq 0)$ be a random walk with $S_0 = 0$ and iid increments $(X_n: n \geq 1)$ defined by $P(X_n = 1) = p$ and $P(X_n = -1) = q$. find the Prob. $(S_n = x)$ at time n .

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(d) Show that Brownian Motion is a martingale

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Qn 2.(a) Let $(S_n: n \geq 0)$ be a random walk with $S_0 = 0$ and iid increments $(X_n: n \geq 1)$ defined by $P(X_n = 1) = p$ and $P(X_n = -1) = q$. If T is the exit time from $[-a, b]$, show that $P(S_T = -a) = b/(a + b)$.

(10 marks)

(b) Find $E(T)$ if T is as defined in part (a) above.

(10 marks)

Qn3.(a). Let B_t be standard Brownian Motion. Consider stochastic differential equation

$$dX_t = 2tX_t + t dB_t$$

Find the solution to the equation above.

(20 marks)

Qn 4.(a) A particle moves in a line according to a standard BM, B_t , what is the expected Position at time $t = 6$, what is the variance at time $t = 6$.

(10 marks)

(b) The price of a commodity moves according to BM $X(t) = \sigma B(t) + \mu t$ with variance term



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$\sigma^2 = 4$ and drift $\mu = -5$, Given that the price is 4 at time $t = 8$. What is the probability that the price is below 1 at time $t = 9$

(10 marks)

Qn 5. (a) Let $Y_1, Y_2, \dots$ be a nonnegative iid random variables with $E(Y_n) = 1$. Show that

$X_0 = 1, X_n = Y_1, Y_2, \dots, Y_n$ is a martingale

(10 marks)

(b) Let S_n be an asymmetric random walk with $p > \frac{1}{2}$ and $T = \inf\{n, S_n = 1\}$. Show that

$S_n - (p - q)$ is a martingale use this property to show that $E(T) = \frac{1}{2p-1}$

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MAIN CAMPUS

MAC 819: LINEAR MODELS IN STATISTICS

Date: 13th August, 2018

Time: 11.00 - 2.00pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions



QUESTION ONE (30 MARKS)

(a). Consider a linear regression model.

(i) Show that the model can be written in matrix form as

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$$

where $\underline{Y}$, $\underline{\beta}$ and $\underline{\varepsilon}$ are vectors of order $n \times 1$; $(k+1) \times 1$ and $n \times 1$ respectively,

while X is a matrix of order $n \times (k+1)$.

(4marks)

(i) Define the Least Square Estimator of the parameter vector $\underline{\beta}$ and show that it is given by

$$\hat{\underline{\beta}} = (X^T X)^{-1} X^T Y$$

(4marks)

(b) Consider the following data on two variables, Y and X.

Y	X	Y	X	Y	X
95	96	72	89	35	0
80	77	66	47	50	30
0	0	98	90	72	59
0	0	90	93	55	77
79	78	0	18	75	74
77	64	95	86	66	67

Assuming that Y depends on the X independent variable

(i) Give the appropriate linear regression model, stating all assumptions made.



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77	64	95	86	66	67

Assuming that Y depends on the X independent variable

(i) Give the appropriate linear regression model, stating all assumptions made.

(ii) Compute the model parameter estimates, and give the prediction at $X=1$.

(iii) Determine the 95% Confidence limit for the model regression parameter

(8 marks)

(c) Determine the 95% prediction interval for the Y variable at $X=80$, assuming the results in 1(b).
(4 marks)

(d). Let $\underline{Y} = \beta_0 + \beta_1 X + \varepsilon$ where ε is the error term. Suppose that the Least Square Estimators of β_0 and β_1 are , say $\hat{\beta}_0$ and $\hat{\beta}_1$ respectively.

Show that

(i). $E(\hat{\beta}_0) = \beta_0$

(ii). $E(\hat{\beta}_1) = \beta_1$

Determine

(lii). $\text{Var}(\hat{\beta}_0)$

(iv). $\text{Var}(\hat{\beta}_1)$

(v). $\text{Cor}(\hat{\beta}_0, \hat{\beta}_1)$

(10 marks)

QUESTION TWO(20 MARKS)

(a). Show that if $E(\underline{Y}) = X\underline{\beta}$ and $\text{Cor}(\underline{Y}) = \sigma^2 I$ then the least square estimators $\hat{\beta}_j$,
 $j = 0, 1, \dots, k$, have minimum variance among all linear unbiased estimators.

(10 marks)

(b). Using part (a) or otherwise, show that if $E(\underline{Y}) = X\underline{\beta}$ and $\text{Cov}(\underline{Y}) = \sigma^2 I$, then the best linear unbiased estimator of $\underline{a}^T \underline{\beta}$ is $\underline{a}^T \hat{\underline{\beta}}$ where $\hat{\underline{\beta}}$ is the least square estimator of $\underline{\beta}$.
(5marks)

(c). Does the results in part (a) rely on the distribution of the random vector $\underline{Y}$?
Comment
(5marks)

QUESTION THREE (20 MARKS)

7	10	4	7
8	7	6	10
9	8	6	11
10	12	6	9
11	11	8	15
12	14	8	13

(a). Show how the data can be modeled by a regression model given by

$$\underline{Y} = X \underline{\beta} + \underline{\varepsilon} \quad (5\text{marks})$$

(b). Compute the Least Square Estimate of $\underline{\hat{\beta}}$. (5marks)

(c). If $\text{Var}(\underline{\varepsilon}) = \sigma^2 I$ is known, calculate $\text{Var}(\underline{\hat{\beta}})$. (5marks)

(d). Calculate the estimate of the estimator S^2 , defined in question one (b). (5 marks)

QUESTION FIVE (20 MARKS)

Suppose we fit the model $\underline{Y} = X_1 \underline{\beta}_1^x + \varepsilon^x$ when the correct model is

$$\underline{Y} = X_1 \underline{\beta}_1 + X_2 \underline{\beta}_2 + \underline{\varepsilon} \quad \text{with} \quad \text{Cov}(\underline{Y}) = \sigma^2 I$$

(a). Show that

(i) the Least Square Estimator for $\underline{\beta}_1^x$ is

$$\underline{\hat{\beta}}_1^x = (X_1^T X_1)^{-1} X_1^T \underline{Y} \quad (3 \text{ marks})$$

(ii) Compute the model parameter estimates, and give the prediction at $X=1$.

(iii) Determine the 95% Confidence limit for the model regression parameter

(8 marks)

(c) Determine the 95% prediction interval for the Y variable at $X=80$, assuming the results in 1(b).
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(d). Let $\underline{Y} = \beta_0 + \beta_1 X + \varepsilon$ where ε is the error term. Suppose that the Least Square Estimators of β_0 and β_1 are , say $\hat{\beta}_0$ and $\hat{\beta}_1$ respectively.

Show that

(i). $E(\hat{\beta}_0) = \beta_0$

(ii). $E(\hat{\beta}_1) = \beta_1$

Determine

(iii). $\text{Var}(\hat{\beta}_0)$

(iv). $\text{Var}(\hat{\beta}_1)$

(v). $\text{Cor}(\hat{\beta}_0, \hat{\beta}_1)$

(10 marks)

QUESTION TWO(20 MARKS)

(a). Show that if $E(\underline{Y}) = X\underline{\beta}$ and $\text{Cor}(\underline{Y}) = \sigma^2 I$ then the least square estimators $\hat{\beta}_j$,
 $j = 0,1,...,k$, have minimum variance among all linear unbiased estimators.

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